

Data Warehousing and Data Mining.

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CO2 ASSESSMENT TOOL-2

CSA1613

Q1 (a) Data Mining:

Data mining is the process of finding useful patterns and knowledge from large datasets.

Two Tasks:

- classification
- clustering

KDD steps:

- selection
- cleaning
- Integration
- Transformation
- Data mining
- pattern evaluation
- knowledge presentation.

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Role of preprocessing:

- Improves data quality
- Removes errors
- Increases accuracy.

(b) Data cleaning: Removing incorrect or incomplete data.

Data Quality Issues:

- Missing values
- Noisy data.

Missing values: Replace with mean/median/mode or remove records.

• Noisy data: use binning, regression, or clustering.

Example:

80, 75, 3, 85 $\rightarrow$ Missing value = 80.

20, 21, 22, 50, 23 $\rightarrow$ 50 is noise; after smoothing: 20, 21, 22, 23.

Q2(a)

Data Integration: It is the process of combining data from multiple sources into a single, consistent dataset.

Two challenges:

- Data Redundancy
- Data Inconsistency

Handling challenges:

- Redundancy: Remove duplicate records using keys and normalization
- Inconsistency: standardize formats and validate data.

Q2(b)

Covariance Analysis: It measures how two variables change together.

Two covariance Formulas:

1. population covariance:

$$\text{cov}(x, y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$

sample covariance

$$\text{cov}(x, y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

Interpretation :

- positive covariance : variables increase or decrease together.
- Negative covariance : One increases while other decreases.

Q3 (a)

Given Data :

stock A	stock B
2	5
3	8
5	10
4	14
6	14

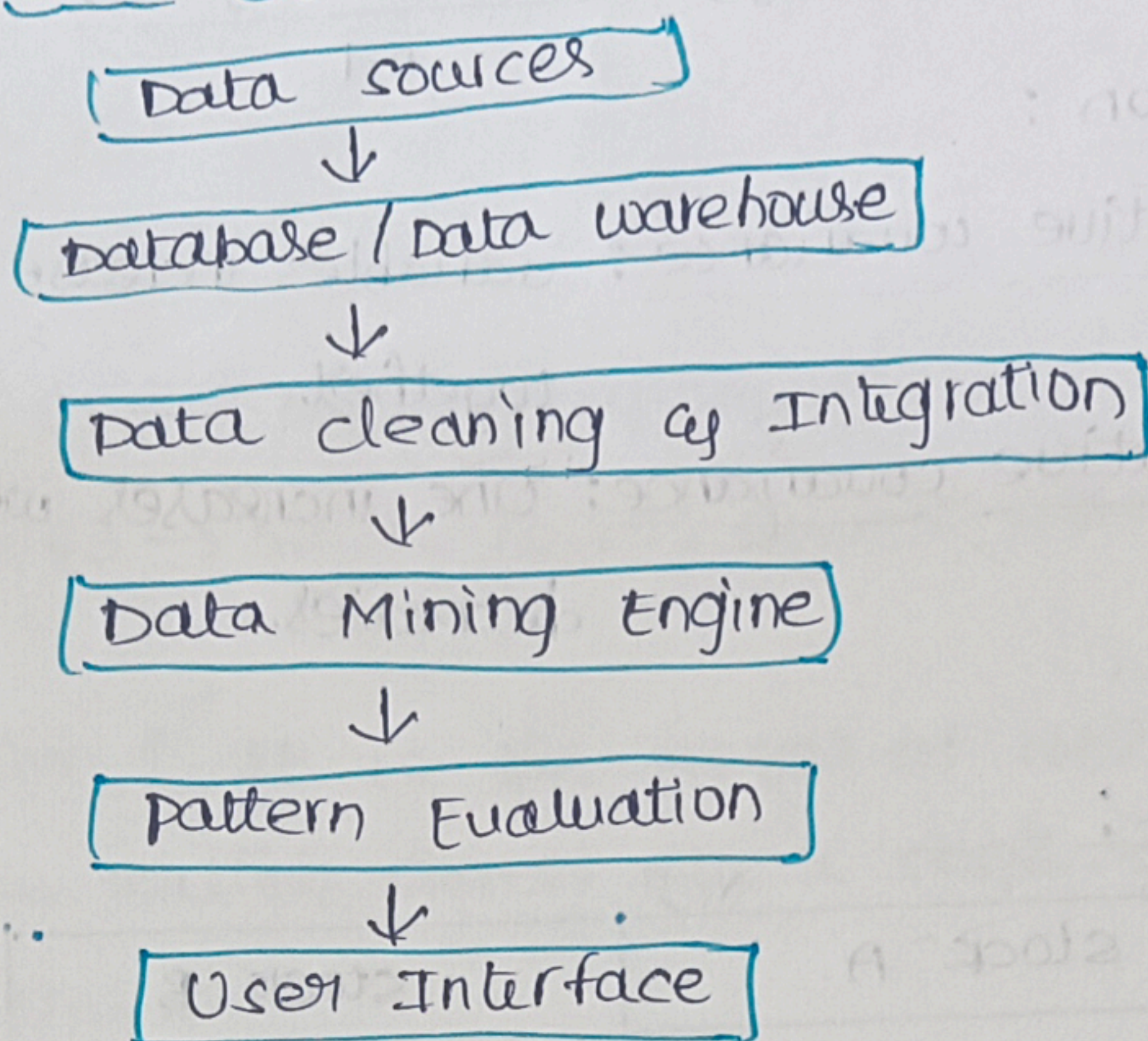
Answer :

As the values stock A increase, the values of stock B also increase. Therefore, the stocks have positive covariance.

conclusion : The stock prices rise together and are not independent.

Q3. (b)

Architecture of a Data Mining system.



Components:

- Data sources: collect raw data
- Database / data warehouse: store integrated data
- Data cleaning & Integration: Improve data quality
- Data Mining Engine: Finds patterns using algorithms.
- Pattern Evaluation: selects useful patterns.
- User Interface: Display the results to users.

Q4(a)

Given:

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = 6.2$$

$$\text{Mean}(Y) = 5.8$$

using the covariance formula,

$$* \text{Cov}(X, Y) = \frac{\sum (X - \bar{X})(Y - \bar{Y})}{N}$$

Covariance = -0.36

Answer: $\text{cov}(x, y) = -0.36$

4(b)

Major steps in Data preprocessing

- Data cleaning
- Data integration
- Data transformation
- Data reduction
- Data discretization

preprocessing pipeline:

collect data



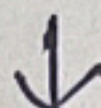
Data cleaning



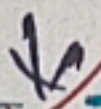
Handle Missing values



Remove outliers



Normalize Data



Data Mining.

* Clean data gives better Mining Results.

5. (a) Equal - Frequency (Equi - Depth) Binning

- Total values = 27
- Number of bins = 3
- values per bin = 9

Bin 1 : 13, 15, 16, 16, 19, 20, 20, 21, 22 → Mean = 18

Bin 2 : 22, 25, 25, 25, 25, 30, 33, 33, 35 → Mean = 28

Bin 3 : 35, 35, 35, 36, 40, 45, 46, 52, 70 → Mean = 44.

Answer: smoothed values are 18, 28 and 44 for the 3 bins.

(b)

Data discretization:

converts continuous data into intervals (bins)

Importance:

- Reduces data complexity
- Improves analysis
- Removes noise

Comparison:

Equal - width	Equal - Frequency
same bin width	same number of values
simple	Balanced data

Impact: Equal - frequency is better for uneven data, while equal - width is suitable for uniform data